

ActInf Textbook Group ~ Cohort 7 ~ Session 3

(Modeling & General Discussion) ~ 7/29/2024

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all right greetings everyone thank you for joining so we're in this first up of the third weeks for the cohort it'll be a bit of a grab bag so we'll see where it goes but a few people have written some first questions um and then we'll look at some prior questions see what else comes up all right so first I'd be interested in examples on how to train an active inference model what what would anyone say on this how is training an active inference model similar or different than potentially other kinds of models um hi uh I can say uh so um I was thinking like um there should be a world model which would be trained on the um on minimizing the variation of free energy uh as a loss objective so I mean I'm assuming that can be for for from from an uh like an RL perspective or something like from some collected trajectories that can be trained but but for the expected free energy uh I'm I'm not so confident like how H how how that can be deployed yeah thank you Frasier I would just I mean I've only successfully made maybe like two or three active inference models that are like non-toy examples but one thing that I found particularly helpful was or or Salient was that you had to really think about the different time scales of things that we're going to be training so you know I guess in in a machine learning context training refers to the updating of parameters of a model so you know and and as was just uh said by is it three I think it's three we need our our model to embody uh the environment in some way so that's going to be parameterized there G to be a bunch of parameters and we need to tweak those parameters in a machine learning setting um that happens at a much slower or at least a slower time scale than the actions that we need to decide to emit at each time point if it's a a discrete time model let's say so there's kind of two questions there's the question of what actions do we emit based upon expected free energy based upon maybe just vfe and then there's the question of okay what how do we also minimize VF with respect to our parameters so those two questions Loom large at least in my mind when I'm thinking about training an active agent cool yeah well there's there's a lot of pieces to to bring out here um one note is that variational basian methods like the variational

auto encoder are also trained or updated using variational free energy minimization so the concept of using the evidence lower bound to to um bound and approximate surprise which is equivalent to maximizing model evidence that's not a new modeling trick from active inference part of the interesting thing though is bringing the perceptual and the action selection process into the unified objective and then using the evidence lower bound on that so so there's like some I mean it it will be cool to kind of pull out from the conversations and papers and get to the vend diagram so that's that's one thing is training a model based upon free energy minimization is something you'll find in a variety of machine learning methods um an interesting difference in terms of of at least like in terms of how chapter six differs uh from some other machine learning Pathways like nowhere in chapter 6 The the recipe for modeling does it say anything like go find a data set of positive examples and then tune your model so that it recapitulates the maximum performance on the positive data set so it's kind of interesting is like prior data can have a different role in the situation of training an active inference model so sometimes in machine learning you have a you have a basically a priori prior setting of a general or a large topology of the model like here's the architecture of the neural network and the layers or I'm training a 7 billion parameter Transformer architecture so you kind of have a large generic function approximator where the nodes are essenti not intended to have any semantic meaning and then during the training phase you pump cats and dogs with labeled examples and then you have the data set of cats and dogs and then you have some held out data set of cats and dogs and you try to improve performance on that so in in this situ in the machine learning situation you you have a kind of generic model architecture usually with many many parameters like thousands to billions and then positive examples are pushed through that Network so that the weights come to embody patterns in in the data in contrast in at least in terms of how chapter 6 lays out the process the basian graphs that are constructed in active inference are smaller and usually more interpretable where the the nodes have distributions that are kind of like based upon how the problem was carved out to be like you have a node that's explicitly based upon the cat or the dog um so that's kind of a difference is rather than walking into the situation with a sort of General highly parameter uh uh architecture and a data set here as chapter 6 lays out is you have a bit more of like a systems modeling approach so that that's one important difference but how how do you um update the the parameters um that's pretty General because there's a lot of different parameters to train Let's uh look at so coming to like chapter 7 this is just focusing on the downstairs sense making piece of what later action will get plugged into so here prior D hidden State s Through

Time State transitions B each state is emitting observations you can think of that as being like the hidden state multiplied by the ambiguity Matrix leads to an observation distribution that's the generative direction or the synthetic data generating direction or you could take an observation multiply it by the ambiguity Matrix and kind of run it back to reconstruct an uncertainty distribution over hidden States so that's part of the flexibility depending on the situation you might say well we know the observations without error like they might be noisy but we measured them without error and then we want to learn some other parameter or we know the hidden States through time but we want to infer the prior so this kind of gets into which variables are fixed and which variables are learnable the way that a distribution is fixed in a Bayesian setting it's like it has a hyper prior so a prior on that distribution that's just a spike like the observation was seven so just a spike at seven whereas if it was the observation was seven plus or minus one then that would take in a whole distribution so just to conclude and give a first pass General answer when you're set up the model if a variable is entered as fixed and with a kind of infinite Precision on it it's not learnable whereas distributions that have like actual breadth those parameter updates can be updated within that range of the distribution what what what there there's a lot of other pieces that was totally not a complete answer what are any thoughts or questions from there I guess most broadly you have to choose a model you have to choose a particular model something like a POMDP or or a Gaussian mixture or or something uh whatever it is that discussion applies to your choice of model but you do have to choose a model yeah another just well these are just thinking of broader modeling considerations so it was brought up how do we train on what hasn't happened like with expected free energy so let's look at just a first pass on what the expected free energy calculation entails it's going to take in a policy prior π it's a vector of policies policy the policy space just like in other control theoretic formalizations the policy space is a vector of all the possible policies policies are all the combinations of affordances for a given time Horizon so if we had four affordances up down left right time Horizon of two there'd be 16 policies up up up right Etc we're going to take this policy Vector which is the probability distribution of action it sums to one something must happen and we're going to update the policy prior into the policy posterior policies are going to be sharpened according to their score on this unified objective a good score can be gained from having either a high pragmatic value which is to say that the sequence of expected future observations is aligned with preferences that's utility Andor the policy has a high expected Information Gain or epistemic value which is to say that the KL Divergence about

beliefs about hidden States through time is diverging because of the observations that are expected to come in so different policies might be valuated to have different pragmatic values and different epistemic values there is this question though how do you balance these two components of the expected free energy because just being just having both of them in a unified objective doesn't mean that there's going to be um an Adaptive finessing of this explore exploit type tradeoff so um first like here at the parametric level here's what that updating or training looks like it looks like changing the amplitude of a variable so [Music] um let's see what chapter okay so here we are in chapter 7 it's a rat and a team as it has a preference for the attractive stimulus so um check the script to see whether this is written um in terms of log probabilities Etc but a six in the preference Vector here means that the attractive stimulus is considered e to the 6 times more probable If This Were one and negative one the relative preferences would have all remained the same If This Were 600 you know you can imagine it would be exponentially more so the way that pragmatic value gets scaled is with scaling the amplitude of the C Vector so you you play around with with you could take the C vector and then just like in the code you could make a series of of C Vector Alternatives like multiply by 0.1 multiply by 0.5 .9 1.1 1.5 2 10 so generate this like Suite of C's bringing you closer to on one hand a flat C like if you said multiply this by one millionth so it's it's basically zeroed out that is equivalent to having a flat preference distrib distribution having a flat preference distribution is equivalent so here's PR now pragmatic is on the first term here having a flat preference distribution is equivalent to Infomax on the other hand you could make your C Vector super massive that would make the epistemic component of the expected free energy negligible which would essentially look like utility uh maximizing control so one lever that you have is modifying the pragmatic value that's pretty straightforward because you're just doing a vector lengthening or shortening and then the other way that you can rebalance the scales is with epistemic value um if your parameters are not learnable like if they're isn't anep if there isn't an uncertainty distribution so you can imagine increasing the Precision on your variables until they're essentially fixed then epistemic value goes down conversely if you were to expand your uncertainty on the variables epistemic value would go up so whether by combination of widening and narrowing your uncertainty distributions to manipulate epistemic value or stretching or shrinking the C Vector to modify the pragmatic value so those are kind of the two knobs that you have to to to manipulate the relative intensities of these two terms but then how do you really select a given parameterized tradeoff so that's a situation where um like going back to the um Tas so this not not the only way to do this but this is just one one

way to do it would we we set up the teams with a given setting now as with any training the optimal parameterization is going to depend on the niche so if we had an example where the food was always on the left then the optimal parameterization is going to be different than if it was switching like a metronome or if it was switching with some autocorrelation function so that's why it does depend on how you set up the niche then you could take let's just say we built five C vectors ranging from a kind of like the 0.1 the 0 five the one the two and the 10 and then five different uncertainty distributions that we could choose from so now we have 25 combinations then we could basically simulate with all 25 of those combinations and then say well which of those mice had the highest performance and says okay well it turns out that given this Niche setup having this C Vector look like this and having the uncertainty distribution look like that gave the highest outcome that's again interesting to consider because it wasn't the case that we had like any positive data set examples so it's a really interesting opportunity to Define these models and then train and tune them in ways that don't necessarily rely on specific past data sets but you can use this sort of synthetic simulation and um method to still develop more useful tunings so that's about the e one option there just being to do parameter sweeps over over possibilities Fraser yeah I was just going to recapitulate that it's so like for you to be able to have an active inference AG per se you have it seems like you have to be able to entertain counterfactual you know possibilities about what would be the case if you didn't an action so you have to be able to perform actions and you have to be able to evaluate the suitability of those actions and the way we do that as with the EFA let's say um you're you're gonna have to have some kind of means of of rank ordering your choices that you make at points in time um that's that's kind of the the ontology of of what you do when you do training let's say of this kind you can do you can do variational free energy minimization but you're not necessarily doing active inference if you're doing that you kind of have to have this um counterfactual uh entertaining of of trajectories into the future it seems so yeah okay let's look over an RX and fur Okay it is figure 1.2 or is it or is it 2.1 oh it should be 2.1 is 2.1 yeah sorry oh no it's it was a blurring the line what table is in what document um so like let's just look at training in the situation of this um this is not an action model this is just an observation model so this is like the listening to music first part of chapter 7 prior beliefs on the frog and the Apple then likelihood model that's going to update based upon an observation here in this case jumping updates to a posterior belief distribution now in in spirit slash um in mathematics this is exactly what happens with updating the action prior into the action posterior here it's just we're talking about a belief prior into a belief posterior conditioned upon

an observation but this is conceptually what happens like let's let's just say this was I have a 10% chance of turning left and then a chance of turning right 90% And then I do G and then update that policy distribution to look more like this and then sample the most likely one or I sample from this distribution so that's the kind of General concept of of updating a distribution but this one's not about action this isn't even an active inference model this is just basian inference yeah yeah this is just basan inference but we can we can say well what what would be the here there would be two things that that might be learned if we were to kind of build this example out you could learn your prior that could come from just simply the prior posterior like from wherever you left off in the previous cycle you just pick up there with your prior um another method for the prior for learning the prior is learning by counting so learning by counting is is very related to the earn model so in this situation you have an urn like a a jar and there's a certain number of balls of color one and color two so initially you don't know you may not even know how many colors there are or maybe you've been told that there's only the two colors you pull out color one you but you don't necessarily want to only believe that there's color one in there so you'd start with one in each and then you as you make observations um you uh count and then it's like a coin flip like if you just kept on Flipping that coin eventually You' get to 100 and 100 and then if you got to a th and a th that would be a sharper prior around 50/50 whereas in one would be like a looser prior so that's one way to learn beliefs um learning a likelihood model is kind of another topic so it's like that's that's part of the challenge sometimes is the variables with different semantics like the hidden State distributions the a matrix the B Matrix the consequences of actions the priors um on preferences c those all have kind of different ways that you might approach learning them because there's different considerations like what will yield the most performance um parameter value then there are considerations like what is a cognitively realistic learning mechanism another consideration would be to what extent do we want to learn this variable and then fix it for runtime or to what extent do we want this variable to be learnable or updatable even in runtime so in this RX and for um model of the figure 2.1 accurately reflecting the textbook the values are all fixed and given but I think it will be very useful for us to with starting with this example then continuing on what does it look like when these distributions are also allowed to to be learned in different ways and crucially for an active inference agent that has to be by means of action yes thank you Dave are any of the implementations of RX and fur able to take account of a situation where not just the values of the variables but what variables there are need to be changed in order to succeed you don't

where you you didn't know that you have to take temperature into account where uh to distinguish between Apples and frogs you didn't know that you have to hum to intervene in to your uh your exploration where is it do you just assume that you know is that something that we're even trying to do at all where there's genuine Discovery where there's no precedent for what how the model has to be changed or change the dimensionality of not just the number of Dimensions but what dimensions there are actually Kus if you have any thoughts feel free to add um I don't really have thoughts uh behind how this would be attempted I just know uh in the AR in first sense that's been a desire to have such a setup uh for a while as far as I understand um for example in the case of the team Ma I know they had I think it was Magnus codol he had to write specifically code and hardcode the structure I think uh based on his needs and I I do understand how he expressed the need at the time to have been able to uh you know uh to rely on uh structured learning to to have that but instead he had to uh provide a custom implementation at that point but I do believe there's an action in the orx in3 development team to take care of this structur approach to learning in the future yeah that'll be that'll be great maybe in the worst case you just have to do something like uh completely jump out and do genetic algorithms something and entirely outside the Paradigm if you're if you're lucky you can simply when you discover relevant dimensions of actual action or sensation just say oh well that was in my in the totality of my prior I just didn't know it none of the values are set just off uh go back can change your the dimensionality the priors and then recalculate but who knows that that that's what that's where you get in some really interesting psychology and really interesting logical techniques thank you I I do know that at the end end of um Magnus kol oh sorry uh Dimitri bv's PhD thesis which is kind of the founding document of ARS INF fur at the end of that he kind of gestures to like other problems now that we want to be able to attack and one of them was this issue of model uh structure learning learning the actual structure of the the the model in that case it's a graph it's very high problem as far as I understand and no one really knows how to do it yet yeah like a few things so you could have an explicit distribution over like I don't know how many lights are going to be in the room I think it could be from 1 to 10 and then you're then you could have methods that oh it's seven now I'm going to spin up seven a matrices for the seven lights but then that kind of like question of unknown unknowns and being able to in a potentially open-ended way incorporate like new new rows in the variables you already have like oh you could go before you could go up down left right now you can also go a diagonal Direction so that kind of like expanding variables you have and then also like oh now you have a new type of affordance or you have a

new sensory modality that um at at least in terms of the examples we have in open question um another quote that I think is kind of just a plant a footnote this is from uh Toby St Clair smi's uh thesis he differentiates between the structured learning and the learning structure so learning structure is about the former extending the process of learning to a structured setting whereas the latter learning the underlying structures themselves it's a kind of playful but interesting question like are we learning the structure of the patterns in the world with a fixed structure internally that's one side of the coin and the other side would be given the structure of the patterns in the world how can the actim model potentially modify its own structure to accommodate those patterns more easily RX infer hopefully provides the sandbox to experiment with those things because of the possible open-endedness of being able to combine different nodes um I don't think any of our examples from kass's work or from the reactive Bas team they but Fraser rovas correct me if if this is not the case but like they have a fixed topology in runtime it's not that the graph has a certain connectedness and then something occurs like a policy decision or some kind of signal and then the graph changes but and then like Dave mentioned genetic algorithms like there there's so many other pieces like the parameter sweep mentioned earlier but like parameter sweeping isn't necessarily part of active inference it's not part of the particular partition it's not the physics of action and perception it's just a computational science method so in practice a variety of methods come into play ranging from parameter sweeps parallelization AB testing so those all come into play to kind of tune or or give a bigger view on the active inference generative model so sometimes you are just interested in building one generative model other times it's like even for just for fun and um speculative realism even then you still may want to generate a sequence of generative models um just like if you were doing linear aggression you would make from simple on through like you have a um a family from the most possible interactions and including all the variables to like the simplest model and then there' be the combinations in in between so there's something kind of analogous to to to cross calibrating the performance of different models that have that have different structural elements now there was a presentation uh might have been a guest stream and I suspect it was in 2020 where somebody was was addressing these using a formal setting uh where graphs combine and split in response to particular kinds of failures do you remember that an ecological treatment of graphs I I do remember Bert D of the bias lab saying that this idea of uh 40 40 St Factor graphs they allow for a relatively robust model in the sense that you can take out nodes and you know the reactive message passing will just kind of self-organize around that um I don't know if that's actually been demonstrated

yet with with ar in particular I think maybe that's more of a theoretical um benefit so far uh but yeah I'm not aware of actual examples of that just yet there there's several other um I think it was the supervised there's several other approaches to to structure learning so first off again a Toby paper this is more on the category Theory here um this is in terms of structured learning because it describes agents with structured interfaces for learning like computer apis so that's super exciting but now on the learning structure side um there's earlier work 2020 Smith at all and then there's a more recent work um which is is quite interesting here okay the the notation and the letters get a little funny and interesting because like here the transition probabilities are encoded in C but in the textbook we're just discussing like C is the preference on observations but here it's not so it just you know see that um what happens in this paper is uh it's in the context of amnest handwriting data and there are basically these um there's an unknown unknown and not preset number of styles and then there's this uh question which is how many styles should we given our task is to categorize these digits into 0o through nine how many styles should we spin up so in the extreme case there's only one style of zero one style of one one style of two that's just just like a big dis big circle around the points in in in uh space on the Other Extreme like every single example is its own style so both the one style and the every single one has its own style those are on two extremes of a structure learning parameterization how many styles are there and so then the question is how do you find that sort of prto optimal Bas optimal number of styles that actually provide you positive classification ability and so that's sometimes approach from this idea of of on one hand adding a latent factor and seeing if there's variability that that latent factor can explain and then testing um you you using um BAS model selection okay here's my model with n Styles here's n plus one style is that extra style punching above its weight or if it's diminishing returns then the smaller model's better so that's the kind of like add a latent factor see if it helps and then move forward whichever one's better and then the other side of that is like the pruning which is kind of like the sleep or the synaptic pruning in that situation you say okay given the N parameters I have which one would be like the first to go and then you remove the least informative of the parameters that you have so that has led to some interesting work probably more not open source than open source on systems that are able to in in different kinds of ways add latent factors to explain and explore deeper patterns in like observation data or like in the consequences of action and then cyclically for example like in a sleep wake wake type situation add parameters during wakefulness to explain more variance remove parameters that are the least informative during sleep to promote model sparcity and interpretability and

simplicity another thing just to note since the question was like about training is chapter nine is where the um the topic of connecting to empirical data is brought in before chapter n it's like all kind of like we proposed this model like we've just a priori specified this model and then we ran the model forward in a generative direction to generate synthetic data the reverse direction is let's just say we already had a data set so we already have a data set of people typing and eye movement or of the mouse in the Maze or the particle moving or whatever it is so we already know what data we want to connect from um so then we we still use the same generative model such that we we know that we want these observations to connect up to the kinds of observations here like we want to have the location observation at every time step so kind of you could pre-register that model or you could build it after the data were collected um and then what's called Model inversion is so here is distribution of the parameters given the observations and actions and then that is being um flipped it's proportional that's the kind of infinity sign to here's the probability of the parameters given the model and then the probability of the actions given the observations in the model and so then this is reconstructing latent parameters Θ in this kind of General linear regression framework against the observations conditioned upon the observations so when we're generating synthetic data you can think about that as a distribution of observations conditioned upon parameters like I'm just saying that the room is um 30 degrees and now I'm going to generate noisy thermometer data so that's like generate me thermometer observations conditioned upon the hidden State s the other way to do an inference the inverted model would be tell me what you can say about s the true temperature and the room conditioned upon these observations that I'm empirically getting so those are kind of the two directions of using a basian model and um I think there's a lot of good work to be done with making examples that do that that kind of features just coming online in pmdp and it's there in RX and furb but we don't have any larger scale or even medium examples but those would be some cool things to look into it it's it kind of returns to chapter 6 like why are we building this model like if we wanted to study how are the relationships between observations and hidden States learned well then we would have a distribution on a or we'd select from multiple a matrices but once we kind of open up all the possible directions of learning like including preference learning like how does the mouse know that it wants the the food versus the aversive of stimulus or how does it come to know the consequences of its actions from a movement pers perspective that that makes the space of the model larger and larger and larger combinatorically whereas the the case where all the parameters are fixed so there's like not really learning per se there could still be real-time

inference but just it's it's not doing any deeper learning that's like the that's like the plug and chug special case point and then for every time you open up learning on a given variable um like first you have a quantitative learning like just updating the values in the cells then you can think more of like structural learning like the a matrix was 2 x two and now it's going to be 2x3 or it's going to be 3x3 and then you can get even into um more kind of severe structure learning like what about a second a matrix or what about another modification on a base graph each of those steps from fixed quantitative tuning structural tuning within a fix topology opening up the topology that's open-ended each of those could be like large increases in the amount of sweeping that you have to do so if you want to make a playful example with open-endedness you can go in that direction um if you had more of a uh if you had a question that defined your pragmatic value such that you wanted to get some utility out of your model at some point you would probably choose which distributions you want to fix like I I'm just going to say it prefers sugar and it avoids poison because I'm not asking the question how does it come to prefer sugar but then if you were asking how does it come to prefer sugar you would want to make that distribution learnable because that's what would address that question but if the question was given that mice like sugar how do they learn um associative um you know between the color of the room and the shock of the foot or something like that like so the the question from the outset sometimes shapes what you situate as learnable or not yeah FR sure I just I just wanted to say with respect to the significance that that discussion has to modeling uh a lot of us you know I'm quite interested in in you know building models of of of this um the the relevance would be you know you you kind of want to hardcode as much as possible not worry about the topology maybe initially not worry about the multiplicity of the a matrices and just kind of do the most uh truncated strict beian inference you can on the most uh you know cut down version of the problem and then and then maybe iteratively up um expose parts of the the model say okay now now we've got our frog jumping let's say or I think that classifies frogs versus apples but you know what if we need to say determine uh also what we want to classify you know frogs versus app versus oranges versus mice um how do you determine that you want to even classify frogs and apples that's that's another action then you're GNA have to build in so that that introduces a huge amount of complexity so to to avoid that if you're just interested in creating a first step in a model the best thing to do is just kind of ignore those problems and hardcode it um essentially I was just just thinking for for the relevance that would have to to modeling efforts yeah that's a great Point like it makes me think about um here like we could just draw a circle around each of these bar charts and this is

kind of like our basian graph kind of you know these are the variables that we're considering it's almost like that's an island coming out of the ocean or or you know choose choose a different way to think about it but it's like if we wanted to explicitly learn the prior we would have another node which is our prior on frogs and apples is it 0. 1.9 plus or minus .05 or is it 0.19 plus or minus 0.1 and then that would be the learnability of this prior and then you could say well how did it come to believe that it was plus or minus .05 on frogs and apples and then it's like so the Buck has to stop somewhere in practice for a particular model but depending on what is useful or epistemic for you you might pull back multiple layers Dave yeah there's actually quite a lot of data that's real world data that's the analysis of which is can be relevant to what we're talking about talking about uh Behavior shaping from animal trainers and then the formalization of that in Skinner style training and uh bubbling up against what that what what kinds of learning what kind of teaching you actually can efficiently explain is a nice a nice problem on itself there certain you just you just cannot readily explain in scenario set up but there's a lot that you really can yes so figure 91 where the experimenter's choice of which stimuli to provide which test questions to send etc those are observations for the mouse and then what the mouse does in terms of selected actions that's the observations for the experimenter this is kind of the cool thing with with the basian methods is you could pre-register or do a statistical power sweep what trains of experimental stimuli under the generative model as I understand it what trains of experimental stimuli would be differentiable with to from each other or or contrastable or not so like an example of this that's that's very um empirical is in fmri so the fmri signal the Bold the brain oxygen related signal that friston and others have studied with SPM for many years so that signal happens on a time scale of several seconds so they have a graphical model that includes the time scale of several seconds and then they can simulate out well if we had two stimuli and they were alternating at the time scale of half a second we wouldn't be able to dis to to determine the the relative efficacy of these two stimuli because they'd be too convoluted by this like 3 second response time on the other hand if you had the the observations alternating every minute it would be too slow you'd only get one change Point detection per minute so then it's like well what time scale should the observations be alternating on to derive the highest statistical power then even with that calculation there's further considerations like oh but then if the the person cognitively understands oh they're alternating every 16 seconds so there's always many considerations in practice um but that's kind of the the fun of the iterated modeling is this back and forth with the synthetic propositions and the empirical data if it's a situation where there is empirical data

I mean the nice thing that it's it's Beijing inference all the way down on all of these things so we kind of know what we're doing in terms of I mean in some sense um we're still doing belief and inference on model structure we're still doing belief and inference on you know the A matrices and then versus the parameters and so on yes unified approach to um handling and updating the variables whether they're uh beliefs about State beliefs about action beliefs about observations Etc um and and to kind of again bring it back to the very beginning we could be actually building from the inside these active inference models or like in figure 9.1 this could be a Transformer or a neural network so also these models could be used to describe and map cognitive architectures or systems are things that aren't active inference things per se but still have interactions across interface so sometimes you're actually building it other times just observing it and then building something to describe the observations which would kind of be like training a uh machine learning or reinforcement learning model on language or on something like that and and that could be used to generate language so again that has a lot in common with other methods it's just that this is a very flexible phrasing of that problem um okay well so at least this was sort of a um General slm modeling discussion but if there are topics that people like want to explore I would highly recommend slash ask put put the questions on the page or in the table before you can leave a comment to like make sure that that they're seen um and just kind of frontload what you want to ask and that will help like bring other people in and and make make uh a fun learning time okay any um last questions I I guess for chapter two where uh we're opening up for just any any uh sailing discussions at all about any parts of chapter 2 you know it's not like it's gonna have to be a modeling discussion or anything like that next session yeah we can do any correct it'll it'll just be we'll do two weeks a chapter two and then we'll see if we want to have another what we want to do in in three weeks from now cool cool okay I'll stop the recording thank you